RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251801040-2025-AIDS

B2

Started on	Tuesday, 15 September 2026, 8:46 PM
State	Finished
Completed on	Tuesday, 15 September 2026, 8:51 PM
Time taken	5 mins 1 sec
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
long long countways(int n){
    if (n==0||n==1||n==2){
        return 1;
    }
    long long w0 = 1;
    long long w1 = 1;
    long long w2 = 1;
    long long current_w = 0;
    for (int i=3;i<=n;i++){
        current_w = w2+w0;
        w0=w1;
        w1=w2;
        w2=current_w;
    }
    return w2;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 1;
    }
    printf("%lld\n", countways(n));
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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